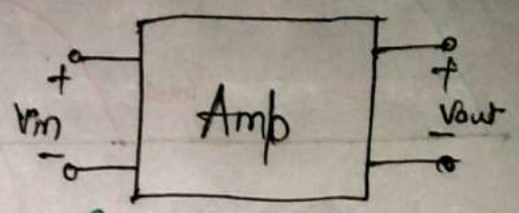
Lecture No. 13

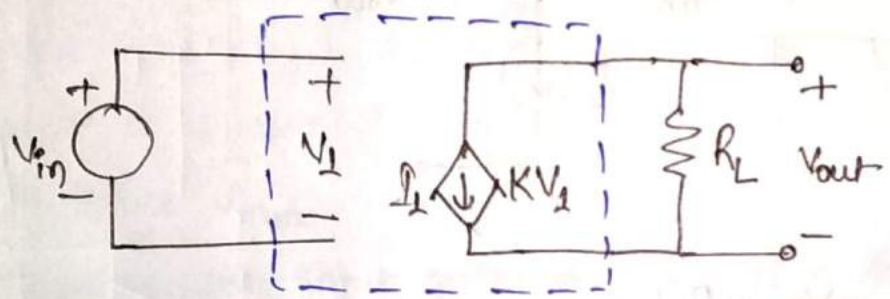
- What's an amplifier?

→ Amplifier is an black box which perhaps receive a voltage and generate a voltage.

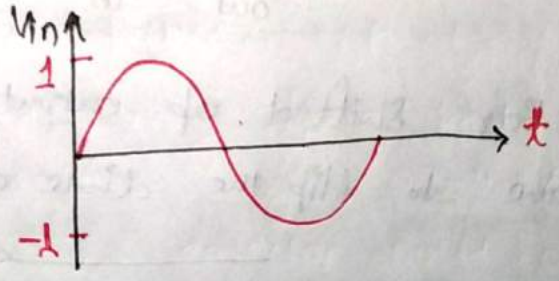


As in figure, if we give a sinusoidal input to the amplifier then it generates a voltage same as input but in greater amplitude.

- Let's build an amplifier



$$\frac{V_{out}}{R_L} + KV_1 = 0$$



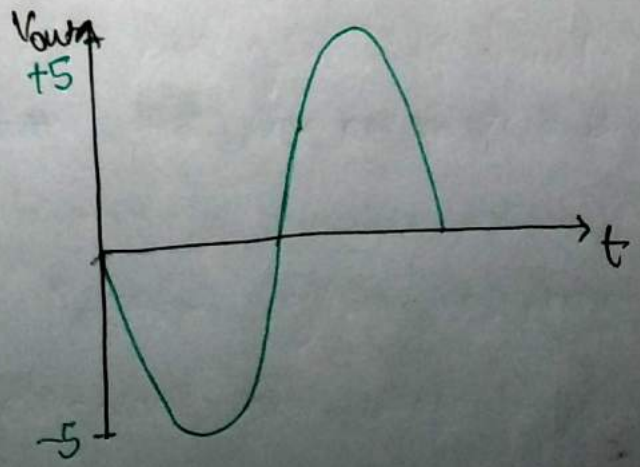
Since $V_1 = V_{in}$ therefore,

$$\frac{V_{out}}{R_L} + KV_{in} = 0$$

$$\Rightarrow \quad V_{out} = -KV_{in}R_L$$

Say $-KR_L = -5$ then

$$V_{out} = -5V_{in}$$



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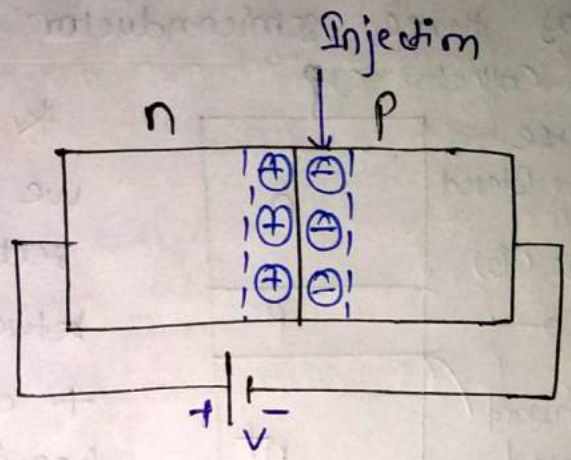
• Since dependent sources are not available in market so we have to find alternative of it which has the function same as dependent source.

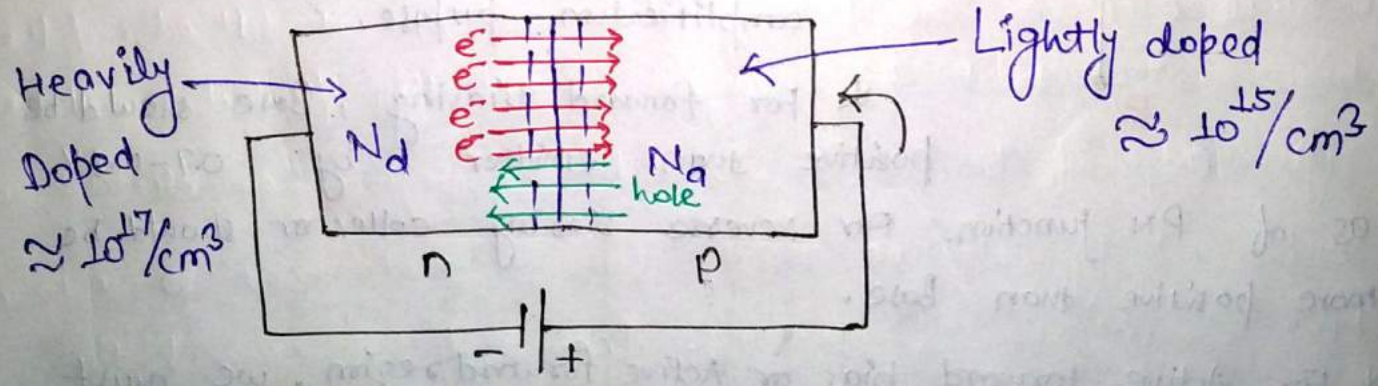
Observation:-

• Concept of carrier injection :-



If we inject electrons on the junction in p-side then we know electric field is present at the junction in direction from left to right then when we inject electrons, these electrons are pushed by electric field to the n side which collects all electrons and these electrons flow to the +ve terminal. As long as we inject electrons then the electrons are pushed to n side always.

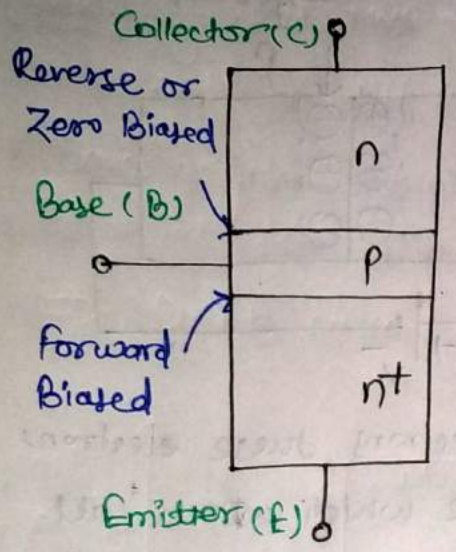
• Effect of a Symmetric Doping :-



→ If we have heavily a symmetric doping means for example high doping level at n side and low doping level at p side. We have lots of electrons at n side but we don't have so much holes at p side so lots of electrons go to p side for flowing

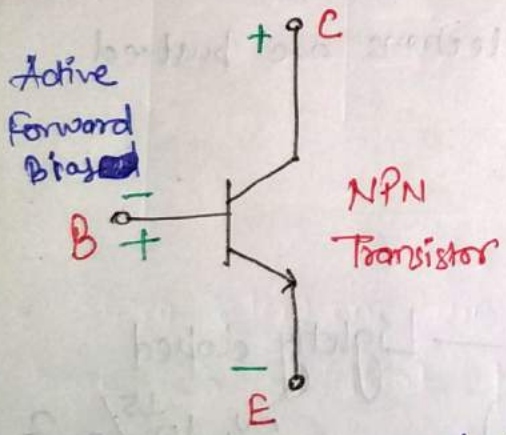
Current and few holes go to n side for flowing current.

Structure and Symbols:- * The Bipolar junction transistor has three semiconductor sections therefore three terminals.



* This device is not symmetric i.e. we can not flip it because bottom section is heavily doped. For distinguish between the sections we just put + on n to denote that it is heavily doped.

* The arrow placed at emitter distinguish between Pnp and npn transistor.



* Forward biased junction between Base & Emitter and reversed biased junction between Collector and base used for amplification purpose.

* For forward biasing, Base should be positive than emitter by 0.7-0.8V as of PN junction. For reverse biasing collector should be more positive than Base.

* For Active forward bias or Active forward region, we must have Forward biased at Base & Emitter and Reverse or zero biased at Collector and Base.

* Base region is very thin, Emitter region has more area than Base and Collector region has larger Area than other region.

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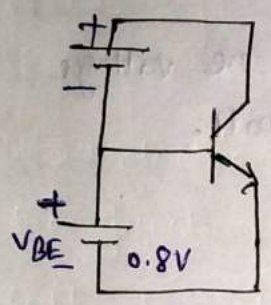
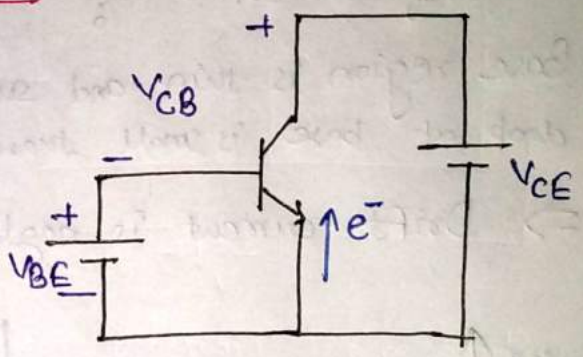
* Emitter is very heavily doped for the purpose of emitting charges, Base is lightly doped and collector is heavily doped than Base and lightly ~~doping~~ doped than Emitter.

* Operation of Bipolar Transistor:-



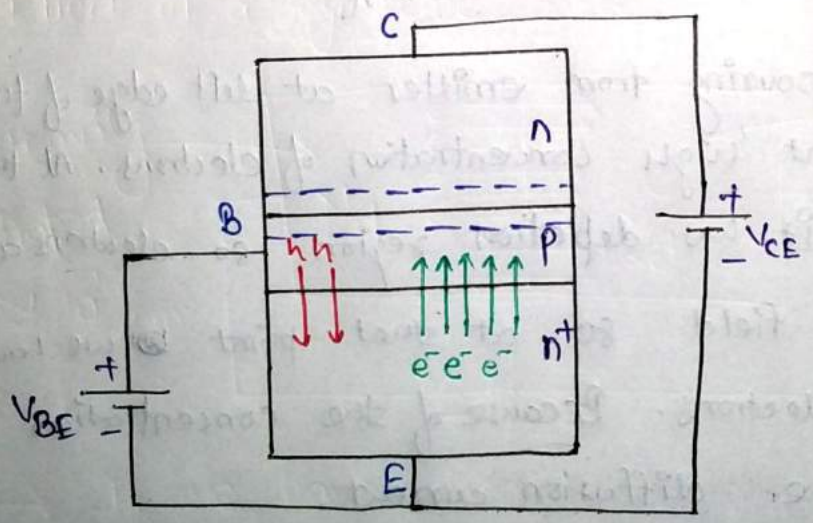
But this cannot amplify more so this configuration is not used.



For reverse or zero biased of CB junction,

$$V_{CB} = V_{CE} - V_{BE} \geq 0$$

$$\Rightarrow \boxed{V_{CE} \geq V_{BE} = 0.8V}$$



* Since we have lots of electrons at Emitter and few holes at Base, these electrons has two routes i.e. Base & Collector.

* The electrons prefers that route which has less resistance. According to the carrier injection at junction electrons are collected by n region at collector and absorb by battery. That's why it is different than PN-junction because in reverse biased C-B region carries current.

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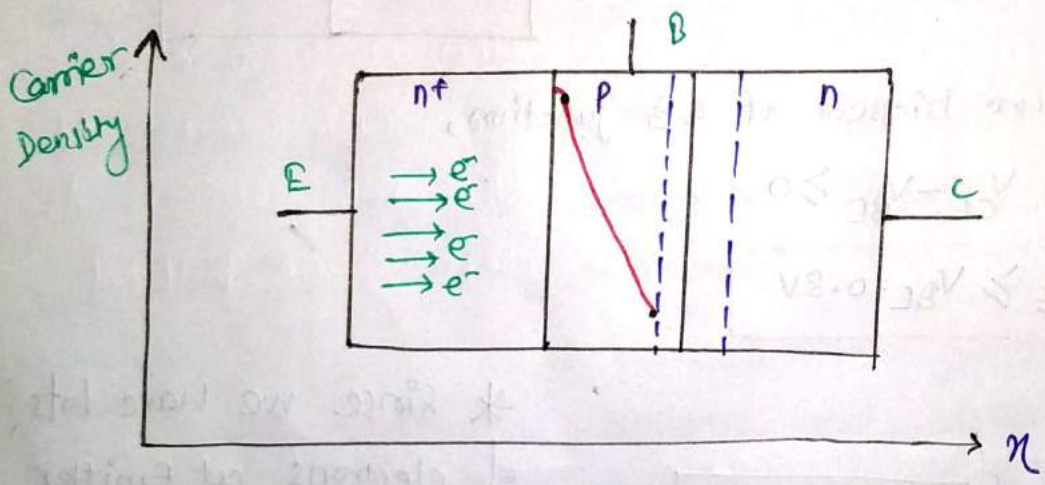
* Since electrons are flowing from emitter to collector so the conventional current associated with positive charges flow from collector to emitter.

Que: By what Mechanism do electrons flow through the base: drift or diffusion?

Base region is thin and sustains little voltage so the voltage drop at base is small therefore electric field is small.

⇒ Drift current is negligible.



- Since electrons are pouring from emitter at left edge of the base so at that point high concentration of electrons. At the right edge of base, it has depletion region so electrons are swept out by electric field so at that point we have low concentration of electrons. Because of the concentration difference we have diffusion current.

Que: What happens if $V_{BE} \uparrow$?

If $V_{BE} \uparrow$ so current increases so electrons increases and holes also increases so we can say that collector current increase.

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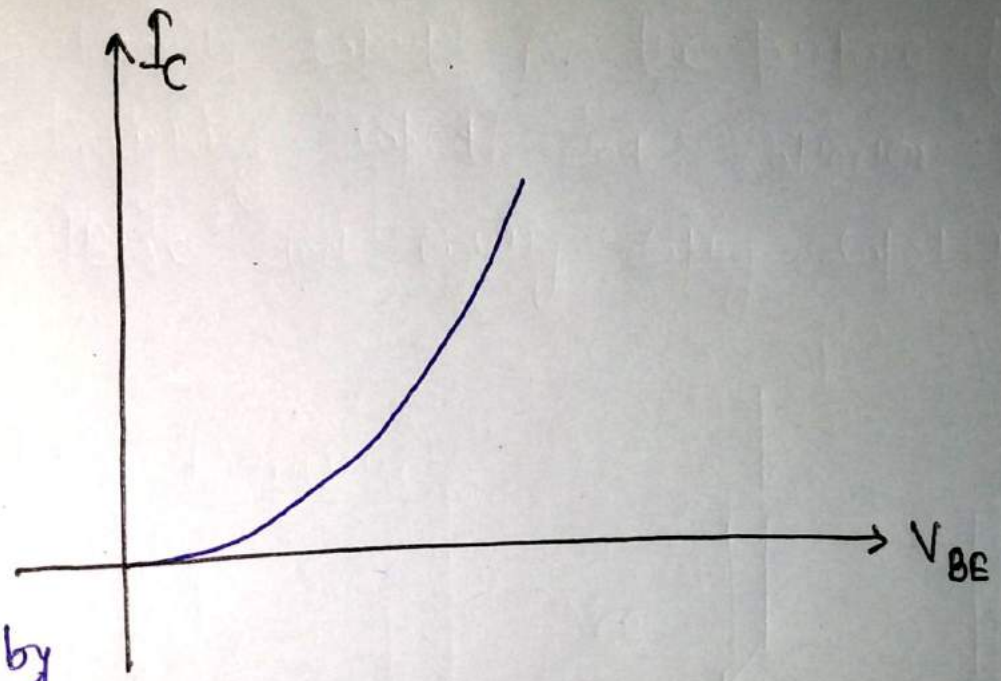

Collector current is related to V_{BE} by

$$\boxed{I_C = I_S\left(\exp\frac{V_{BE}}{V_T} - 1\right)}$$

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